

Environmental evidence annex

ACP-2019-43 (Bournemouth FASI-South) — Runway-26 arrivals, A26-ESE design envelope

Independent supporting evidence for the Brockenhurst Aviation Action Group (BAAG) Stage 3 consultation submission. Prepared from public flight data, 2023–2025. Not affiliated with the airport.

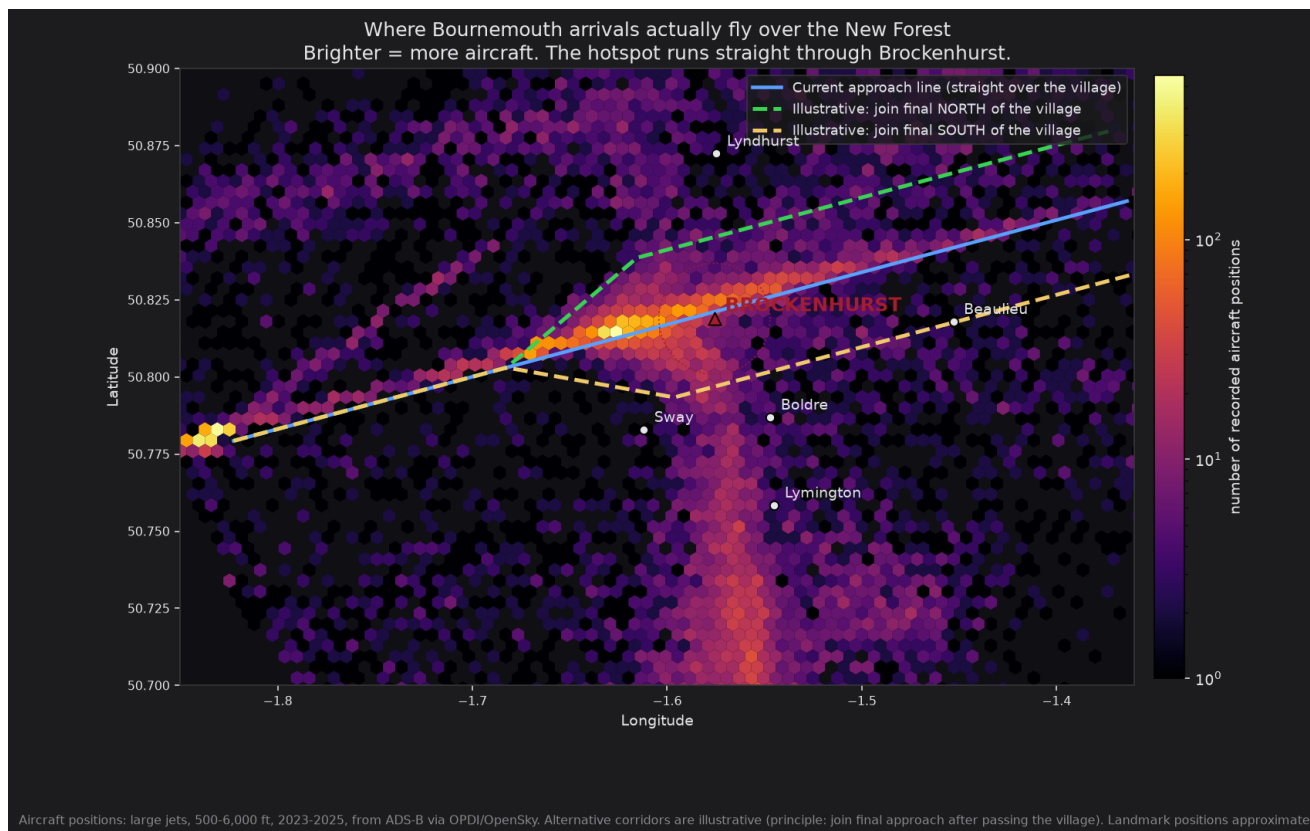
Purpose. This annex supplies the **environmental baseline evidence the airport's own engagement material does not contain**. The 3rd-engagement deck (Cyrrus, ACP 2019-43) defines the runway-26 arrival options **only as lateral track "swathes"** — it makes **no assessment of altitude, noise, Brockenhurst, or the New Forest National Park**. Using three years of the aircraft's own broadcast data, this annex quantifies what is actually happening over Brockenhurst on the **A26-ESE** arrival, and puts numbers behind BAAG's three design demands.

1. The concern, in one line

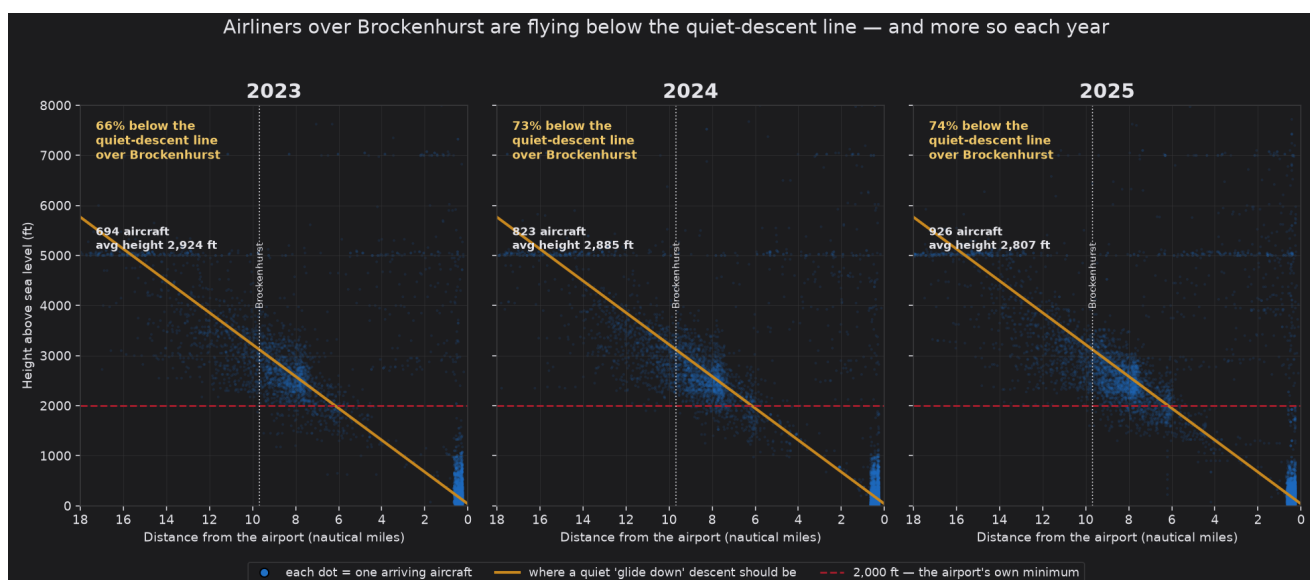
The **A26-ESE** design envelope carries runway-26 arrivals **directly over** Brockenhurst. **The "Do-Minimum" option** replicates today's tracks using precise satellite navigation (PBN/RNAV) — **which would** concentrate **today's scattered** overflights into a **single, permanent, low-altitude line** over the village and National Park: a "noise highway." The evidence below shows the line already exists, is flown too low, and is getting worse.

2. The environmental baseline the ACP omits (measured, not asserted)

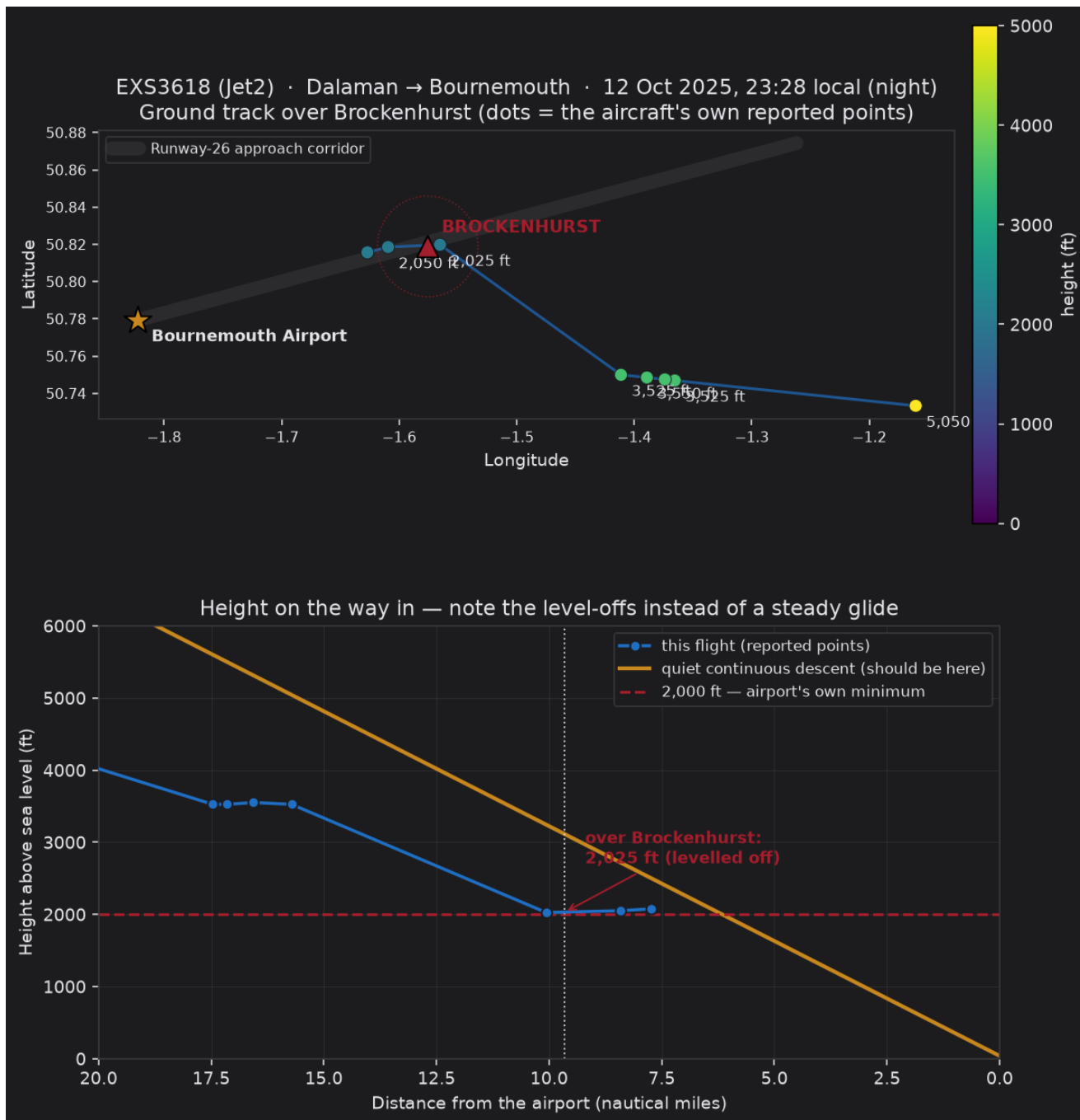
2.1 Brockenhurst is directly under the A26-ESE track. ~76% of airliner arrivals approach over the village (runway-26 operations). The density map shows the concentration running straight through Brockenhurst — the "noise highway" as it is flown today.



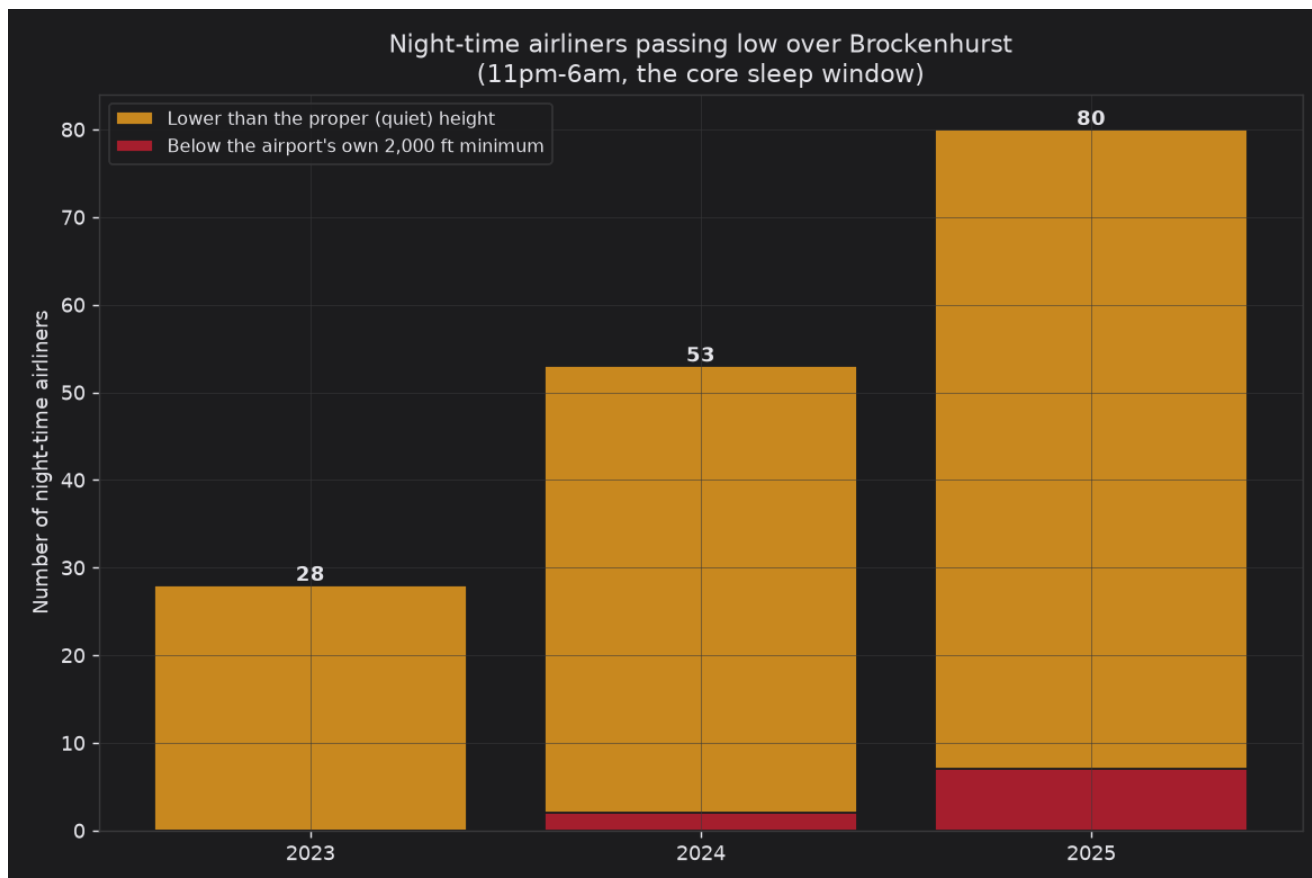
2.2 They are flown lower than a continuous descent. A standard 3.0° continuous descent would place aircraft at **~3,116 ft** over Brockenhurst. Measured reality for airliners: **average ~2,855 ft, median ~2,725 ft**. The share **below** the continuous-descent height rose **66% → 73% → 74%** (2023→2025), and the average is **falling** each year while the number of aircraft **rises**.



2.3 They step down and level off — the high-thrust noise event. A compliant continuous descent has **no level flight**. Level-offs over the village (each one an application of engine power) rose **63 → 95 → 157** per year. Worked example — a Jet2 arrival, night of **12 Oct 2025**, levelled at **~2,050 ft directly over Brockenhurst at 23:28, after also levelling at ~3,500 ft: a stepped, engine-on** profile, built entirely from the aircraft's own recorded positions.

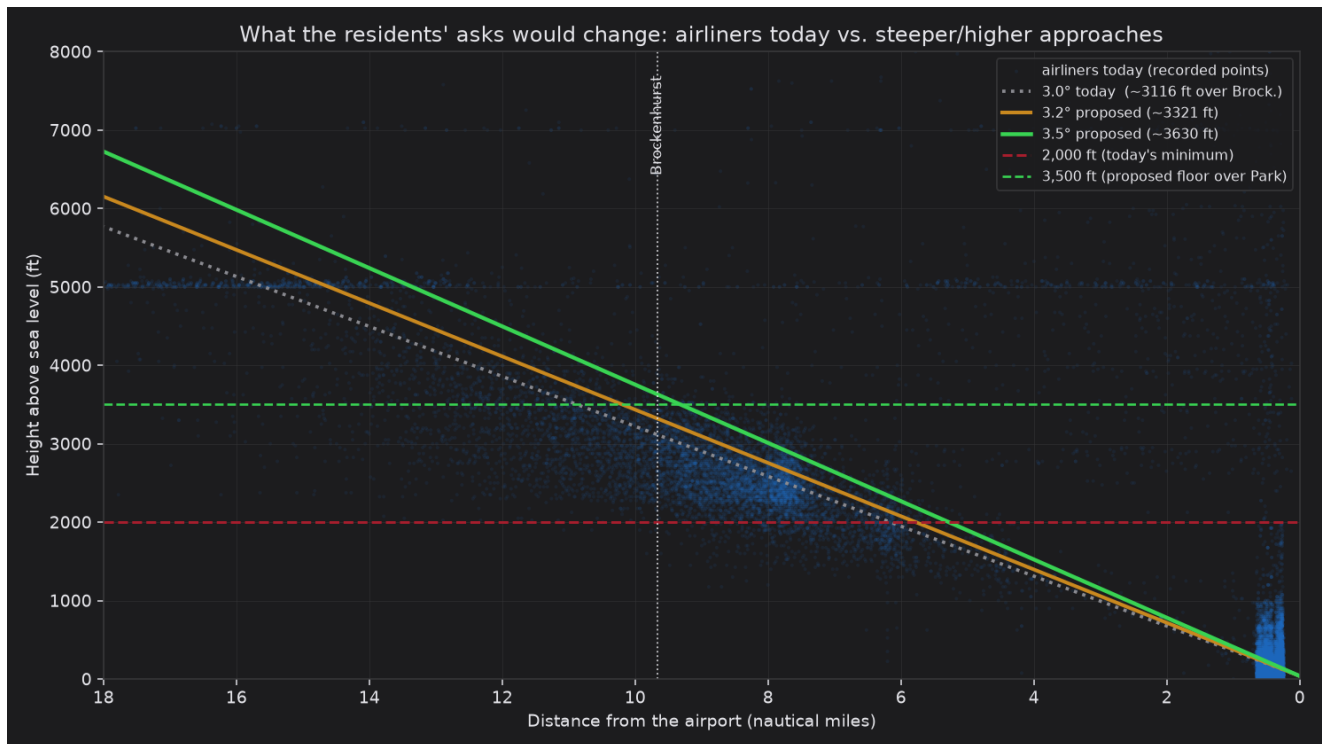


2.4 It is worst at night. Night-time airliners over Brockenhurst rose
263 → 332 → 583 (2023→2025); those **below the airport's own 2,000 ft minimum**
 rose **0 → 2 → 7**.



3. Evidence for BAAG's three design demands

Demand A — steeper approach (3.0° → 3.2°/3.5°). A steeper glidepath keeps aircraft higher over the Park. From our geometry: over Brockenhurst, 3.0° ≈ 3,116 ft, 3.2° ≈ 3,320 ft, 3.5° ≈ 3,630 ft. **Against today's ~2,855 ft average,** that is **+470 ft (3.2°) to +775 ft (3.5°)** — before counting the removal of the low level-offs. The figure shows today's aircraft (blue) sitting **below** both proposed slopes.



Demand B — higher minimum height over the Park (3,500 ft floor). Today's average over Brockenhurst (~2,855 ft) is **~650 ft below** a 3,500 ft floor, and a substantial share are below even the existing 2,000 ft minimum. A hard 3,500 ft floor over the National Park, combined with **no level-offs**, is the single largest noise reduction available and is consistent with the airport's existing **night rule** (establish no closer than 8 miles / not below 2,500 ft).

Demand C — lateral offset south of the village. The density map (2.1) shows both the concentrated over-village track **and** that many aircraft already join from wider out — so an offset is operationally realistic. Technical precision to protect the demand: the **final ~4-5 miles must align with the runway** and cannot be offset; the offset therefore applies to the **intermediate/joining segment (Brockenhurst is ~10 miles out), with aircraft establishing on final after** passing the village. PBN makes such a defined offset precise and repeatable.

4. Measured against the airport's own commitments

The above is not our standard — it is the airport's own:

- **Section 106 (planning permission 8/07/0065):** enforce Continuous Descent Approaches "wherever possible." (BAAG to confirm exact wording.)
- **UK AIP, EGHF AD 2.21:** a "continuous descent... at all times," staying "as high an altitude as practical," joining "without level flight," plus a night rule (2130-0630) of **8 miles / 2,500 ft**.
- The data in §2 shows these are routinely not met on the A26-ESE arrival.

5. Data, method and honest limitations

- Source: aircraft's own **ADS-B** broadcasts, 2023-2025, via **OPDI**

(EUROCONTROL Performance Review Unit) and **OpenSky Network** — independent of the airport, **and** corroborating **the airport's own WebTrak**.

- ~**97%** of arrivals are captured. Precise over-village heights come from a **representative sample** (~1 in 5 of overflying arrivals), so **all counts are** minimums — **the true figures are** higher.
- We do **not** claim every low flight breaches a rule; some have air-traffic reasons. The case is the **consistent, worsening pattern** across thousands of flights.
- Full method, definitions and a flight-by-flight dataset are available on request (documents: Methodology; Data, sources & definitions; Excel flight list).

Prepared by Brockenhurst residents from public data to support ACP-2019-43 Stage 3. Not affiliated with, or endorsed by, Bournemouth Airport.